Comparative Evaluation of YOLOv5s, YOLOv8s, and YOLOv11s for Traffic Object Detection

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***Abstract*—Real-time object detection plays a vital role in the advancement of intelligent transportation systems and autonomous driving technologies. This study presents a comparative analysis of three YOLO (You Only Look Once) versions—YOLOv5s, YOLOv8s, and YOLOv11s—trained from scratch using a unified dataset and comparable training settings. Focusing on the traffic object detection problem, the research investigates the architectural evolution and performance improvements of these models across different generations. A dataset constructed by combining samples from COCO 2017 and the Vehicle Dataset for YOLO, covering three traffic-related classes: car, bus, and motorcycle. Each model is evaluated using standard accuracy and efficiency metrics to reveal trade-offs between detection performance and computational cost. The findings provide insights into YOLO's architectural progression and help identify the most effective version for real-time traffic detection applications.**

***Index Terms*—Real-Time Vehicle Detection, YOLOv5, YOLOv8, YOLOv11, Computer Vision, Model Comparison**

I. Introduction

Real-time object detection has become a fundamental component of intelligent transportation systems and autonomous vehicles. Detecting vehicles and other dynamic traffic elements accurately and efficiently is essential for safe navigation, congestion monitoring, and automated decision-making.

Among numerous detection frameworks, the YOLO (You Only Look Once) family stands out for its real-time inference capability. Over successive versions, YOLO models have improved in accuracy, architectural design, and computational efficiency, making them ideal candidates for deployment in resource-constrained or latency-critical environments. However, most comparative studies of YOLO models rely on pretrained weights or inconsistent datasets, making it difficult to evaluate their true architectural improvements under comparable experimental conditions. Furthermore, limited research focuses specifically on traffic object detection tasks using balanced and unified datasets.

In this study, we train YOLOv5s, YOLOv8s, and YOLOv11s models from scratch using a unified dataset that combines COCO 2017 [1] and the Vehicle Dataset for YOLO [2], focusing on three traffic-related classes: car, bus, and motorcycle. Each model is evaluated under consistent configurations using standard metrics such as mean Average Precision (mAP), Precision, Recall. The objective is to analyze the architectural evolution and performance trends of YOLO models and to identify the optimal balance between accuracy and efficiency for real-time traffic object detection applications, contributing to safer and more efficient intelligent transportation systems.

II. Related works

Several studies have compared successive YOLO versions to explore their trade-offs in real-time object detection. In Mo et al.[3], conducted a comprehensive comparison of several lightweight YOLO variants, including YOLOv5n, YOLOv8n, YOLOv9t, YOLOv10n, and YOLOv11n, to evaluate their performance in real-time vehicle and pedestrian detection tasks. By highlighting the trade-offs between detection accuracy and efficiency across these lightweight variants, the study provides valuable insights into the suitability of different YOLO model scales for real-time and resource-constrained scenarios. These findings motivated the exploration of model variants in this work and guided the selection and evaluation of lightweight YOLO architectures for the proposed project.

Building upon this, In their study, Zhu et al.[4] propose an enhanced YOLOv11-based model (YOLOv11-ND) and conduct extensive experimental evaluations to analyze detection performance in nighttime urban traffic environments. The authors employ multiple evaluation metrics including Precision, Recall, mAP@50 and mAP@50–95 demonstrating that comprehensive metric-based comparison is essential for assessing both detection accuracy and real-time feasibility. In addition, the experiments are conducted on a dataset comprising 10 distinct traffic-related classes, revealing that detection performance varies significantly across categories depending on object size and sample distribution. The results show that classes with sufficient samples and larger object scales achieve higher detection accuracy, while smaller objects or underrepresented classes suffer from reduced performance.

Kumar et al.[5] investigate traffic sign detection by comparing YOLOv5 and YOLOv8, with a primary focus on the architectural evolution within the YOLO family. Their study highlights how YOLOv8 introduces a more refined network structure that improves feature extraction and object localization compared to earlier YOLO versions. Unlike YOLOv5, which relies on previous-generation backbone and detection head designs, YOLOv8 adopts architectural improvements aimed at enhancing robustness under varying lighting conditions and complex visual patterns. The authors emphasize that these structural changes make YOLOv8 more suitable for traffic sign detection tasks involving diverse sign categories and visual variability. This architecture-focused comparison provides insight into how successive YOLO versions evolve to address limitations of earlier models and motivated the exploration of different YOLO model variants in this project.

In their comparative study on road object detection, Sunitha et al.[6] analyze multiple YOLO architectures, including YOLOv5, YOLOv7, and YOLOv8, with a focus on architectural evolution and suitability for real-time road environments. Their discussion highlights that while YOLOv7 introduces certain refinements over YOLOv5, the overall architectural progression between these two versions remains incremental. In contrast, YOLOv8 incorporates more substantial structural enhancements, such as an improved feature pyramid network, optimized anchor handling, and modular design components that facilitate better multi-scale feature representation. The study emphasizes that these architectural changes position YOLOv8 as a more advanced successor within the YOLO family. Based on this observation, the comparison between YOLOv5 and YOLOv8 emerges as a more meaningful baseline for analyzing architectural impact, which directly motivated the model selection and comparative design adopted in this project.

More recently, Das, Ibrahem, and Fouda [7] conducted "A Comprehensive Review on Real-Time Vehicle and Pedestrian Detection Using YOLO," systematically evaluating YOLO versions from v7 through v11. Their findings revealed that YOLOv11 achieved an mAP of 88% at 45 FPS, outperforming all prior versions through innovations such as sparse attention mechanisms and dynamic inference scaling. The study emphasized YOLOv11's scalability and reliability for intelligent transportation and autonomous driving systems.

While these studies collectively demonstrate the rapid evolution and increasing efficiency of YOLO models, most comparisons were conducted on heterogeneous datasets and under differing experimental conditions, which makes direct performance comparison difficult. To address this limitation, the present study performs a unified and controlled evaluation of YOLOv5s, YOLOv8s, and YOLOv11s using a harmonized multi-source dataset derived from COCO and Vehicle Dataset for YOLO. By filtering and combining these sources into a custom dataset focused exclusively on vehicle classes, the data distribution was standardized and label consistency was ensured. This normalization enables a fair and reproducible assessment of detection accuracy, inference speed, and computational efficiency across all three YOLO versions under identical training and validation conditions.

III. Dataset Building

A. Dataset Sources

The COCO 2017 dataset serves as one of the most comprehensive benchmarks for object detection, containing over 118,000 training and 5,000 validation images annotated with 80 object categories. For this project, only traffic-related classes—namely car, bus, and motorcycle—were extracted to form a focused subset suitable for vehicle detection.

In contrast, the Vehicle Dataset for YOLO provides a more domain-specific collection of high-resolution street and traffic images, featuring various vehicle types under different environmental and lighting conditions. By integrating these two complementary sources, the dataset achieves a balance between the large-scale diversity of COCO and the targeted relevance of the Vehicle Dataset for YOLO. This combination results in a robust foundation for training and evaluating real-time traffic detection models.

B. Dataset Construction

The dataset used in this study was systematically constructed through multiple preprocessing and merging stages to ensure balanced and high-quality training data for traffic object detection. Initially, all relevant classes were extracted from the COCO 2017 dataset, and filtered annotation files were generated to include only the required target categories. These filtered instances were then converted from the original COCO format (x min, y min, width, height) to the normalized YOLO format (class id, x center, y center, width, height) for compatibility with the YOLO training pipeline.

Due to the limited number of validation samples in the COCO subset, the Vehicle Dataset for YOLO was also processed through the same filtering and conversion steps. The two processed datasets were subsequently merged to increase data diversity and improve class balance. As a result, a domain-specific dataset was created focusing on three traffic-related classes: car (label 0), bus (label 1), and motorcycle (label 2).

The final dataset comprises approximately 17,000 training images and 1,200 validation images, corresponding to about 7% validation data. Within the training set, there are roughly 13,000 car, 4,600 bus, and 4,200 motorcycle instances. This composition provides a realistic yet sufficiently balanced representation of traffic environments, where cars naturally appear more frequently than other vehicle types. The dataset exhibits diverse lighting conditions, camera angles, and urban contexts, allowing the trained models to generalize effectively to real-world scenarios and supporting a fair performance comparison among YOLO versions.

TABLE I
Number of Instances per Class in the Dataset

Class	**Instance Count**
Car	43,911
Bus	6,403
Motorcycle	9,023

IV. Model Selection

Three versions of the YOLO (You Only Look Once) family—YOLOv5s, YOLOv8s, and YOLOv11s—were selected for this study to represent distinct generational shifts in the architectural design philosophy and performance optimization of real-time object detection models. Each model embodies a major milestone in the YOLO lineage, reflecting how the community's priorities evolved from lightweight deployment and stable pipelines toward anchor-free prediction, dynamic computation, and more adaptive attention mechanisms.

YOLOv5s, introduced by Ultralytics in 2020, employs a CSP-based backbone and a PANet-inspired neck, offering a reliable balance between representational power and computational overhead. With approximately 7.2 million parameters, it is optimized for GPU-efficient training and straightforward deployment, maintaining high compatibility with diverse toolchains and embedded platforms. Its stability, mature ecosystem, and extensive benchmarking history made YOLOv5s an ideal baseline model, enabling fair performance comparison against more recent architectures.

YOLOv8s, released in early 2023, represents the first major architectural pivot in the YOLO series by abandoning anchor-based detection and adopting a fully anchor-free prediction head. This change eliminates the need for predefined anchor boxes, reduces post-processing stages, and improves robustness in scenarios involving objects with varying aspect ratios—an essential characteristic for traffic scenes where vehicles, pedestrians, and motorcycles exhibit highly diverse shapes. With roughly 11.2 million parameters, YOLOv8s also integrates enhanced feature aggregation modules and decoupled detection heads, enabling faster convergence and improved small-object recognition. It was selected to investigate whether anchor-free detection can outperform anchor-based methodologies when trained under identical conditions.

YOLOv11s, the most recent iteration (2024), extends the architectural evolution even further by incorporating sparse attention mechanisms, dynamic label assignment, and refined feature scaling strategies designed to minimize unnecessary computation. Unlike previous versions, YOLOv11 introduces adaptive gradient flow and more efficient spatial attention, enabling the model to selectively focus on salient regions without sacrificing inference speed. With approximately 10.2 million parameters, it delivers higher mean Average Precision (mAP) while reducing latency compared to YOLOv5 and YOLOv8, despite its larger representational capacity. YOLOv11s was chosen to evaluate the impact of these next-generation architectural enhancements and to provide a contemporary benchmark that addresses a current gap in the literature, as few comparative studies have examined YOLOv11 performance at this scale.

The selection of the "small" (s) variants is intentional. These models retain the full architectural identity of their respective series while maintaining a parameter footprint suitable for training on locally available GPUs. This configuration enables a controlled comparison of architectural innovations—CSP-based design in YOLOv5, anchor-free prediction in YOLOv8, and attention-based optimization in YOLOv11—without conflating results with mere increases in parameter count or GPU consumption. Moreover, the "s" versions were preferred specifically because their reduced memory requirements and computational cost align with the constraints of the local training environment, allowing all models to be trained from scratch under identical conditions without access to high-end cloud infrastructure. Table II summarizes the architectural characteristics and parameter distributions of the selected YOLO variants.

TABLE II
COMPARISON OF YOLOv5, YOLOv8, AND YOLOv11 MODELS IN TERMS OF PARAMETER COUNT

YOLO Version	Model Variant	Parameter Count (Millions)
YOLOv5 (2020)	YOLOv5 (overall)	**7.0–87 M**
	YOLOv5s (small)	7.2 M
YOLOv8 (2023)	YOLOv8 (overall)	**3.2–68 M**
	YOLOv8s (small)	11.2 M
YOLOv11 (2024)	YOLOv11 (overall)	**9.0–258 M**
	YOLOv11s (small)	10.2 M

V. Training

A. Model Configurations

All models were trained from scratch (pretrained=False) to ensure a fair comparison of architectural performance without the influence of pretrained weights. The training was conducted with the same hyperparameters: 100 epochs for YOLOv5s and YOLO8s, 150 epochs for YOLOv11s, batch size = 16, input resolution = 640×640, and the Stochastic Gradient Descent (SGD) optimizer (momentum = 0.937, initial learning rate = 0.01). Data augmentation techniques—random horizontal flipping, mosaic augmentation, and brightness/contrast variation—were consistently applied across all models to improve generalization.

Because the dataset contained class imbalance (with car instances being more dominant than bus and motorcycle), class weighting was applied using the –weights parameter to ensure equal contribution of all classes to the loss function. This adjustment mitigated model bias toward the majority class and improved detection consistency across categories.

All experiments trained on NVIDIA RTX 2060 GPUs under identical conditions. The trained models were evaluated using standard object detection metrics, including mAP@0.5, mAP@0.5:0.95, Precision, Recall and Training and Validation Losses allowing a comprehensive and unbiased performance comparison.

B. Evaluation Metrics

To ensure a rigorous and interpretable evaluation of model performance, a set of widely accepted object detection metrics was utilized in this study. The primary metric employed was the mean Average Precision (mAP), which reflects the overall detection quality across all classes. Specifically, mAP@0.5 measures prediction accuracy at an Intersection over Union

(IoU) threshold of 0.5, providing a coarse yet informative indicator of localization performance. In contrast, mAP@0.5:0.95 averages precision values across multiple IoU thresholds ranging from 0.5 to 0.95 at increments of 0.05, offering a more stringent and fine-grained assessment of detection robustness. Together, these two variants allow for a comprehensive comparison of both loose and strict bounding-box matching criteria.

To further examine class-wise detection behavior, Precision and Recall were analyzed. Precision quantifies the proportion of correctly predicted positive detections among all predicted positives, thereby indicating the model's resistance to false alarms. Recall, on the other hand, measures the proportion of correctly detected objects relative to all ground-truth objects, capturing the model's ability to avoid missed detections. These complementary measures highlight whether a model prioritizes conservative predictions or broad coverage, which is critical in safety-sensitive scenarios such as traffic surveillance.

In addition to accuracy-based metrics, validation classification loss (val/cls-loss) was used to inspect the model's generalization capability during training. This loss function reflects how effectively the model distinguishes between object classes on unseen data; a decreasing validation classification loss indicates that the model is improving without overfitting. Monitoring this metric enables the identification of optimal training epochs and provides insight into model stability across architectures.

Finally, to analyze prediction errors at a more granular level, a Confusion Matrix was constructed using the predicted and ground-truth labels. This matrix visualizes misclassification patterns between classes and reveals common sources of confusion—such as visually similar objects or underrepresented categories. The Confusion Matrix therefore complements aggregate performance metrics by exposing per-class weaknesses and guiding targeted refinement of both model architecture and dataset composition.

VI. Results

A. Training Loss Evaluation

During the training stage of all three YOLO models, the evolution of the loss components—box regression loss (train/box_loss), classification loss (train/cls_loss), and distribution focal loss (train/dfl_loss)—was monitored to evaluate the optimization behavior of each architecture. Across the experiments, all models exhibited a consistently decreasing trend in these losses, indicating stable gradient propagation and the absence of vanishing or exploding gradients. The train/box_loss demonstrated the fastest initial decline, reflecting rapid learning of spatial localization and bounding box refinement. In parallel, the train/cls_loss reduced more gradually, as the models required additional epochs to correctly differentiate between visually similar traffic classes such as cars, buses, and motorcycles. The train/dfl_loss, responsible for fine-grained bounding box distribution learning, converged at a slower pace relative to box regression but ultimately reached lower values, signifying improved confidence and precision in predicted bounding-box offsets. Among the three architectures, YOLOv11s converged the fastest and achieved the lowest overall training losses, benefiting from its refined attention-based design. YOLOv8s followed with stable and smooth convergence associated with its anchor-free detection head, while YOLOv5s displayed the slowest yet steady decline, consistent with its earlier-generation optimization scheme. These observations confirm that all models successfully learned the underlying object representation patterns without instability, while YOLOv11s demonstrated superior optimization efficiency during training.

TABLE III
Training Loss Metrics of Evaluated YOLO Models

Model	train/box_loss	train/cls_loss	train/dfl_loss
YOLOv5s	0.26558	0.21465	0.29873
YOLOv8s	0.19626	0.16705	0.24428
YOLOv11s	**0.18032**	**0.14129**	**0.23014**

B. Validation Loss Analysis

To complement the training loss evaluation, the validation loss curves were analyzed to investigate the generalization capabilities of the models on unseen data. Three core validation loss components were monitored: validation box loss (val/box_loss), which reflects the model's ability to regress bounding box coordinates accurately; validation classification loss (val/cls_loss), which measures the correctness of class predictions; and validation distribution focal loss (val/dfl_loss), which captures the model's confidence and refinement in predicting bounding box distributions. Together, these losses provide a comprehensive view of the detection pipeline's localization and semantic accuracy during inference.

Across all three models—YOLOv5s, YOLOv8s, and YOLOv11s—a consistent and monotonic decline in all validation loss metrics was observed. Notably, none of the losses exhibited late-stage divergence, oscillation, or upward trends, indicating that none of the models suffered from overfitting, despite their differing training durations. Both YOLOv5s and YOLOv8s were trained for 100 epochs, while YOLOv11s was trained for 150 epochs. The fact that YOLOv11s continued to reduce all validation loss components without any degradation in performance confirms that its extended training period did not cause memorization of the training data and instead reinforced stable generalization.

Among the models, YOLOv11s achieved the lowest final val/box_loss, val/cls_loss, and val/dfl_loss values, demonstrating superior optimization efficiency and more accurate bounding box representation. YOLOv8s displayed smooth convergence in all three losses, aided by its anchor-free detection mechanism, while YOLOv5s exhibited the slowest decrease, consistent with its earlier-generation backbone and optimization routines. The convergence behavior across validation loss metrics therefore follows a clear generational trend that mirroring the architectural improvements introduced in successive YOLO releases.

TABLE IV
FINAL VALIDATION LOSS METRICS OF EVALUATED YOLO MODELS

Model	val/box_loss	val/cls_loss	val/dfl_loss
YOLOv5s	1.02421	0.83132	1.04736
YOLOv8s	0.91254	0.71643	0.89471
YOLOv11s	**0.85487**	**0.63982**	**0.83265**

C. Quantitative Performance Evaluation

To comparatively assess the detection performance of the evaluated models, four quantitative metrics were considered: mAP@0.5, mAP@0.5:0.95, Precision, and Recall. These metrics collectively capture both localization accuracy and classification reliability under varying Intersection over Union (IoU) thresholds. Table V summarizes the final validation results obtained at the completion of each model's training cycle. The YOLOv5s model, serving as the baseline, achieved an mAP@0.5 score of 0.7189 and an mAP@0.5:0.95 score of 0.5498, demonstrating a reliable but expected performance level consistent with its established architecture. Its Precision value of 0.7402 indicates moderate confidence in its detections, although the Recall score of 0.6806 suggests occasional missed detections, particularly in classes with limited representation.

The YOLOv8s model improved upon this baseline by adopting an anchor-free detection head, resulting in an mAP@0.5 of 0.7319 and an mAP@0.5:0.95 of 0.5669. This improvement reflects a more robust bounding box regression mechanism capable of handling objects with varying aspect ratios. Additionally, YOLOv8s achieved the highest Recall value among all three models (0.7033), indicating that it successfully detected a greater proportion of ground-truth objects. However, its Precision (0.7351) remained slightly lower than that of YOLOv5s and YOLOv11s, implying a higher tendency toward false-positive detections—a characteristic consistent with aggressive object hypothesis generation typical of anchor-free architectures.

The most recent architecture, YOLOv11s, outperformed both predecessors across nearly all detection metrics. Recording an mAP@0.5 of 0.7476 and an mAP@0.5:0.95 of 0.5823, YOLOv11s delivered the most accurate bounding box predictions, benefiting from its refined feature scaling, sparse attention mechanisms, and dynamic label assignment strategy. Furthermore, the model achieved the highest Precision score (0.7899), indicating that its predictions were not only accurate but also highly selective, with minimal false positives. Although its Recall (0.6817) did not surpass YOLOv8s, it remained competitive and balanced, demonstrating that YOLOv11s retains a conservative detection strategy without significantly compromising overall detection coverage.

Taken together, these results confirm a clear performance progression across successive YOLO generations. YOLOv8 introduces architectural innovations that enhance generalization and detection completeness, while YOLOv11 consolidates these improvements by incorporating more advanced attention-based optimizations to achieve superior precision and overall mAP performance. Consequently, YOLOv11s emerges as the most reliable and accurate model in this evaluation, offering the best trade-off between detection precision, generalization capacity, and bounding box regression quality under identical training conditions.

TABLE V
FINAL VALIDATION METRICS OF EVALUATED YOLO MODELS

Model	mAP@0.5	mAP@0.5:0.95	Precision	Recall
YOLOv5s	0.71890	0.54983	0.74022	0.68065
YOLOv8s	0.73191	0.56689	0.73514	**0.70326**
YOLOv11s	**0.74761**	**0.58228**	**0.78990**	0.68172

D. Confusion Matrix Interpretation

To investigate class-specific detection behavior, the confusion matrices of the three YOLO models were analyzed. Each matrix presents the normalized prediction distribution across four classes—car, bus, motorcycle, and background—revealing how frequently each class was correctly identified or misclassified.

The YOLOv5s model exhibited a tendency to misclassify true objects as background, particularly in the car class, where 0.66 of car instances were incorrectly labeled as background. While the model demonstrated relatively strong performance for the bus class (0.82 correct predictions), the elevated background misclassification indicates that YOLOv5s struggles with dense or visually complex scenes. This behavior aligns with its lower Recall score, confirming that YOLOv5s misses a considerable portion of true objects despite maintaining acceptable class discrimination.

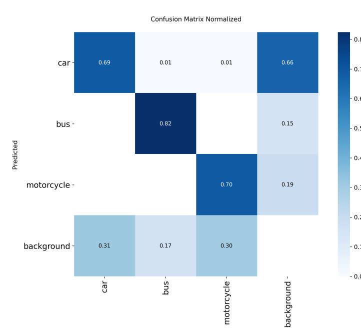

Fig. 1. Trained YOLOv5s Confusion Matrix

In contrast, YOLOv8s reduced background confusion and produced the highest correct detection rates for the motorcycle class (0.72), which is typically more challenging due to its smaller scale and diverse appearance in traffic scenes. The improved separation between vehicle classes and background is consistent with YOLOv8s's anchor-free architecture, which allows the model to detect a wider variety of object shapes. This behavior resulted in the highest Recall value among the three models, indicating that YOLOv8s identifies more true objects, albeit at the cost of slightly increased false positives when compared to YOLOv11s.

The YOLOv11s model exhibited the most balanced error distribution across all classes. Although its car accuracy (0.67)

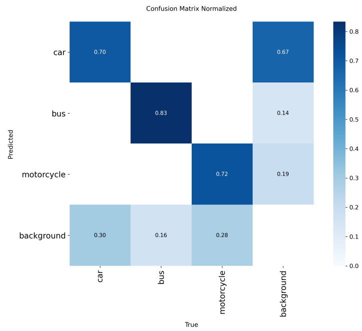

Fig. 2. Trained YOLOv8s Confusion Matrix

was marginally lower than YOLOv8s, the model achieved the lowest bus $\rightarrow$ background confusion (0.12) and reduced motorcycle misclassification compared to YOLOv5s. This indicates that YOLOv11s is more selective in its predictions, avoiding unnecessary detections and thus minimizing false positives—a property also reflected in its superior Precision metric reported earlier. The model's improved class separation supports the efficacy of the architectural refinements introduced in YOLOv11, particularly the sparse attention and refined feature scaling modules.

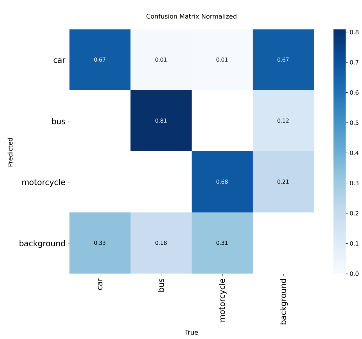

Fig. 3. Trained YOLOv11s Confusion Matrix

Overall, the confusion matrices confirm the quantitative trends established in previous subsections: YOLOv8s excels in broad coverage and object recall, YOLOv5s exhibits limitations in separating objects from background, and YOLOv11s provides the most precise and reliable predictions by minimizing inter-class confusion. These results demonstrate a clear generational progression in the YOLO family, reinforcing the architectural advantages of the newer models for real-time traffic object detection applications.

E. Performance Comparision Summary

The comparative analysis of YOLOv5s, YOLOv8s, and YOLOv11s demonstrates a clear evolutionary improvement across successive generations of the YOLO architecture. While YOLOv5s provides a stable baseline, its higher background confusion and slower loss convergence reveal limitations in complex and dynamic traffic environments. YOLOv8s significantly reduces these issues through its anchor-free detection strategy, achieving the highest Recall and demonstrating superior ability to detect a broader range of objects. However, this comes at the cost of increased false positives, as reflected in its lower Precision relative to YOLOv11s. The most recent model, YOLOv11s, achieves the optimal balance between generalization and selectivity: it delivers the highest mAP scores, the lowest training and validation losses, and the most refined confusion matrix patterns, indicating superior bounding-box regression and class discrimination. Its performance remained stable even after extended training to 150 epochs, exhibiting no signs of overfitting. Collectively, these results position YOLOv11s as the most effective architecture among the evaluated models, providing the best trade-off between accuracy, robustness, and computational efficiency for real-time traffic object detection.

TABLE VI
TRAINING TIME COMPARISON OF YOLO MODELS ON NVIDIA RTX 2060

Model	Epochs	Total Training Time (hours)	Avg Time per Epoch (min)
YOLOv5s	100	10.034	6.02
YOLOv8s	100	10.229	6.14
YOLOv11s	150	15.993	6.40

VII. CONCLUSION

This study presented a comprehensive comparative analysis of three modern YOLO architectures—YOLOv5s, YOLOv8s, and the recently released YOLOv11s—for real-time traffic object detection involving cars, buses, and motorcycles. The models were trained and evaluated under identical conditions, enabling a fair examination of their detection accuracy, generalization behavior, and classification reliability. Experimental results showed a clear generational improvement across versions, with YOLOv11s achieving the highest mAP scores, the lowest training and validation losses, and the most balanced confusion matrix distribution. These findings not only confirm the architectural effectiveness of the newest YOLO family member but also fill a current gap in the literature, where performance analyses of YOLOv11 remain limited.

The outcomes of this research demonstrate that YOLOv11s provides a robust and scalable foundation for intelligent transportation systems, traffic surveillance, and autonomous mobility applications that require high precision under real-time constraints. While YOLOv8s remains a strong candidate for scenarios prioritizing maximum object coverage, YOLOv11s offers the most reliable trade-off between detection accuracy and computational efficiency.

VIII. FUTURE WORK

Future work may extend this study by training full-scale YOLO model variants on larger and more class-balanced datasets to better assess their true representational capacity and generalization ability. Expanding the dataset size and diversity would enable a more comprehensive evaluation of model robustness across varying object scales and scene complexities, while reducing class imbalance effects observed in lightweight configurations. Such an extension would provide deeper insight into the scalability of modern YOLO architectures and their performance potential in real-world traffic monitoring and autonomous perception systems.

TABLE VII
RECOMMENDATION SUMMARY BASED ON EVALUATION CRITERIA

Evaluation Criterion	Best Performing Model
Highest Precision	YOLOv11s
Best Recall	YOLOv8s
Lowest Model Complexity (Parameters)	YOLOv5s
Best Overall Detection Performance	YOLOv11s

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